

## Problem 11.25

A deposit of 10,000 is made into a fund at time  $t = 0$ . The fund pays interest at a nominal rate of discount  $d$  compounded quarterly for the first two years. Beginning at time  $t = 2$ , interest is credited using a nominal rate of interest of 8% compounded quarterly. At time  $t = 5$ , the accumulated value of the fund is 14,910. Calculate  $d$ .

### Solution

During the first two years, there are  $4(2) = 8$  quarterly conversion periods. The accumulation factor associated with the nominal discount rate is therefore

$$\left(1 - \frac{d}{4}\right)^{-8}.$$

From time  $t = 2$  to time  $t = 5$ , there are  $4(3) = 12$  quarterly conversion periods. The corresponding accumulation factor is

$$\left(1 + \frac{0.08}{4}\right)^{12}.$$

Equating the accumulated value at time 5 to 14,910 gives

$$\begin{aligned} 10,000 \left(1 - \frac{d}{4}\right)^{-8} \left(1 + \frac{0.08}{4}\right)^{12} &= 14,910, \\ \left(1 - \frac{d}{4}\right)^{-8} &= \frac{14,910}{10,000 \left(1 + \frac{0.08}{4}\right)^{12}}. \end{aligned}$$

Solving for  $d$ ,

$$\begin{aligned} 1 - \frac{d}{4} &= \left[ \frac{14,910}{10,000 \left(1 + \frac{0.08}{4}\right)^{12}} \right]^{-1/8}, \\ d &= 4 \left\{ 1 - \left[ \frac{14,910}{10,000 \left(1 + \frac{0.08}{4}\right)^{12}} \right]^{-1/8} \right\} \\ &\approx 0.0800949852. \end{aligned}$$

Thus, the nominal rate of discount compounded quarterly is

$$\boxed{d \approx 0.080095 = 8.0095\%}.$$